

Probability and Statistics – for Computer Science

Based on Forsyth, *Probability and Statistics for Computer Science* (2018)

Applied Statistics and Probability for Computer Scientists

Useful Probability Distributions

Chapter 5 — Discrete, Continuous, Normal, and Approximations

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Probability & Statistics for Computer Science

Chapter Outline | What we will cover

Discrete Distributions I
Discrete Distributions II
Continuous Distributions I
Continuous Distributions II

The Normal Distribution
Properties of the Normal
Normal Approximation to the Binomial
Summary

Learning Objectives

- ▶ Recognize when to use each **named distribution**
- ▶ Know the **mean** and **variance** formula for each
- ▶ Understand the **normal distribution** and the 68–95–99.7 rule
- ▶ Apply the **normal approximation** to the binomial, with continuity correction



Section 1

Discrete Distributions I

Discrete Uniform & Bernoulli | The two simplest building blocks

Discrete uniform (Def. 5.1)

$X \in \{a, \dots, b\}$, each equally likely:

$$P(X = k) = \frac{1}{b - a + 1}, \quad E[X] = \frac{a + b}{2}$$

Bernoulli (Def. 5.2)

One trial, success probability p :

$$P(X = 1) = p, \quad P(X = 0) = 1 - p$$

$$E[X] = p, \quad \text{var}(X) = p(1 - p)$$



Bernoulli($p=0.7$) PMF

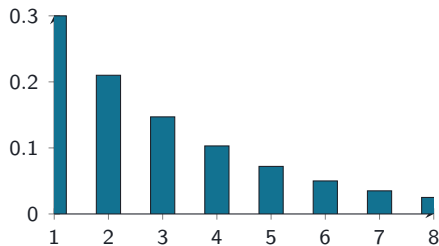
The Geometric Distribution | Waiting for the first success

Definition 5.3

Number of trials until (and including) the first success:

$$P(X = k) = (1 - p)^{k-1}p, \quad k = 1, 2, \dots$$

$$E[X] = \frac{1}{p}, \quad \text{var}(X) = \frac{1-p}{p^2}$$



Geometric($p=0.3$): probability first success is on trial k

Section 2

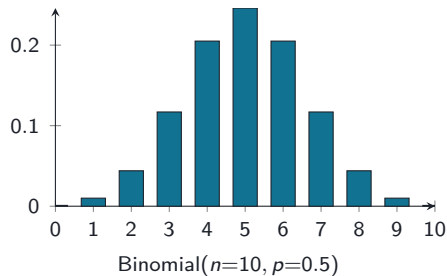
Discrete Distributions II

The Binomial Distribution | Counting successes in n independent trials

Definition 5.4

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

$$E[X] = np, \quad \text{var}(X) = np(1 - p)$$



Multinomial & Poisson | More than two categories; counting rare events

Multinomial (Def. 5.5)

n trials, k outcome categories with probabilities $p_1, \dots, p_k$:

$$P(n_1, \dots, n_k) = \frac{n!}{n_1! \dots n_k!} p_1^{n_1} \dots p_k^{n_k}$$

e.g. counting outcomes over many die rolls.

Poisson (Def. 5.6)

Counts of rare events in a fixed interval, rate λ :

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}, \quad E[X] = \text{var}(X) = \lambda$$

e.g. server requests per second, typos per page.

Section 3

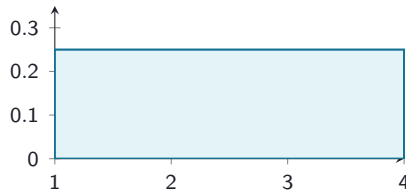
Continuous Distributions I

The Continuous Uniform Distribution | Equal density everywhere on an interval

Definition 5.7

$$f(x) = \frac{1}{b-a}, x \in [a, b]$$

$$E[X] = \frac{a+b}{2}, \quad \text{var}(X) = \frac{(b-a)^2}{12}$$



The Exponential Distribution | Waiting time between rare events

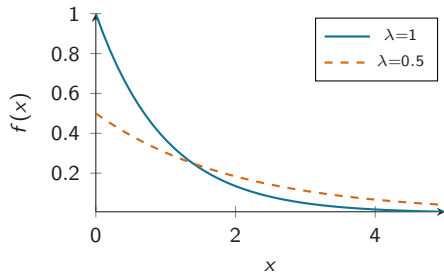
Definition 5.10

$$f(x) = \lambda e^{-\lambda x}, \quad x \geq 0, \quad E[X] = \frac{1}{\lambda}$$

Memoryless property

$$P(X > s + t \mid X > s) = P(X > t)$$

The past wait length does not matter.



Section 4

Continuous Distributions II

The Gamma Distribution | Waiting for the α -th event

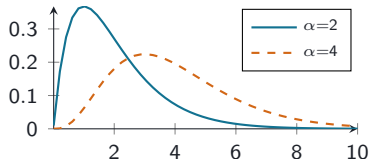
Definition 5.9

$$f(x) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}, \quad x > 0$$

$$E[X] = \frac{\alpha}{\beta}, \quad \text{var}(X) = \frac{\alpha}{\beta^2}$$

Sum of exponentials

The sum of α iid $\text{Exp}(\beta)$ variables is $\text{Gamma}(\alpha, \beta)$ — generalizes the exponential.



The Beta Distribution | Modeling a probability itself, on $[0, 1]$

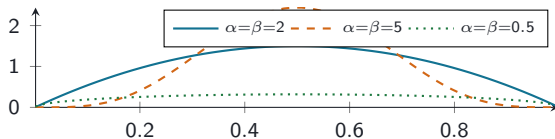
Definition 5.8

$$f(x) = \frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha, \beta)}, \quad x \in [0, 1]$$

$$E[X] = \frac{\alpha}{\alpha + \beta}$$

Why it matters

Natural prior/posterior for an unknown probability p (e.g. a coin's bias) — central to **Bayesian inference** in Chapter 9.



Section 5

The Normal Distribution

Why the Normal Distribution Matters | The most important distribution in statistics

- ▶ Arises naturally as a sum of many small effects
- ▶ Fully described by just μ and σ
- ▶ Foundation of confidence intervals and hypothesis tests

Definition 5.12

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

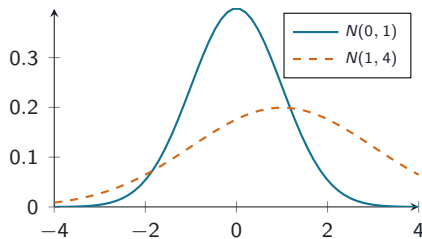
The Standard Normal & Standardization | Every normal is a shifted, scaled standard normal

Definition 5.11: $Z \sim N(0, 1)$

$$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$$

Standardization

If $X \sim N(\mu, \sigma^2)$, then $Z = \frac{X - \mu}{\sigma} \sim N(0, 1)$.

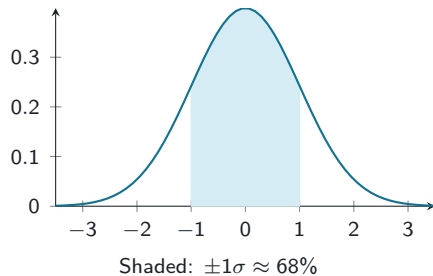


Section 6

Properties of the Normal

The 68–95–99.7 Rule | Almost all the probability lies within 3 std. devs.

- ▶ $\approx 68\%$ within $\mu \pm 1\sigma$
- ▶ $\approx 95\%$ within $\mu \pm 2\sigma$
- ▶ $\approx 99.7\%$ within $\mu \pm 3\sigma$



The Central Limit Theorem (Informal) | Why the normal shows up everywhere

Informal statement

The sum (or mean) of many independent random variables — *regardless of their own distribution* — is approximately **normal** for large n .

Why it matters

Justifies using the normal distribution to approximate binomials, sample means, and measurement errors.

Section 7

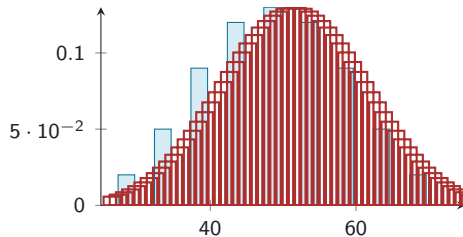
Normal Approximation to the Binomial

When Can We Approximate? | A rule of thumb for large n

Rule of thumb

Use $X \sim \text{Bin}(n, p) \approx N(np, np(1 - p))$ when

$$np \geq 5 \quad \text{and} \quad n(1 - p) \geq 5$$



Binomial bars vs. normal curve overlay

The Continuity Correction | Bridging discrete steps with a continuous curve

The issue

X is discrete (integer steps); the normal approximation is continuous — naive substitution loses accuracy at boundaries.

The fix

Treat $P(X \leq k)$ as $P(X \leq k + 0.5)$ under the normal curve:

$$P(X \leq k) \approx \Phi\left(\frac{k + 0.5 - np}{\sqrt{np(1-p)}}\right)$$

Worked Example | 100 fair coin flips: at most 45 heads?

$$n = 100, p = 0.5: np = 50, \sqrt{np(1-p)} = 5.$$

$$P(X \leq 45) \approx \Phi\left(\frac{45.5 - 50}{5}\right) = \Phi(-0.9) \approx 0.184$$

When the approximation fails

For small n or extreme p (e.g. $p = 0.01$, $n = 20$), $np < 5$ — use the exact binomial or Poisson approximation instead.

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Section 8 Summary

Distribution Cheat Sheet | Mean and variance at a glance

Distribution	Use case	Mean	Variance
Bernoulli (p)	Single yes/no trial	p	$p(1 - p)$
Binomial (n, p)	# successes in n trials	np	$np(1 - p)$
Geometric (p)	Trials to first success	$1/p$	$(1 - p)/p^2$
Poisson (λ)	Rare events per interval	λ	λ
Exponential (λ)	Waiting time	$1/\lambda$	$1/\lambda^2$
Normal (μ, σ^2)	Sums of many effects	μ	σ^2

Chapter Summary | Key takeaways

Discrete world

Bernoulli $\rightarrow$ Binomial (repeats), Geometric (waiting), Poisson (rare counts).

Continuous world

Uniform, Exponential (memoryless), Gamma, Beta — and above all, the **Normal**.

The bridge

The normal approximates the binomial for large n — always apply the **continuity correction**.

Looking ahead

Part III: using samples to **infer** the parameters of these distributions.

Thank you!

Questions & Discussion

Next: Chapter 6 — Samples and Populations